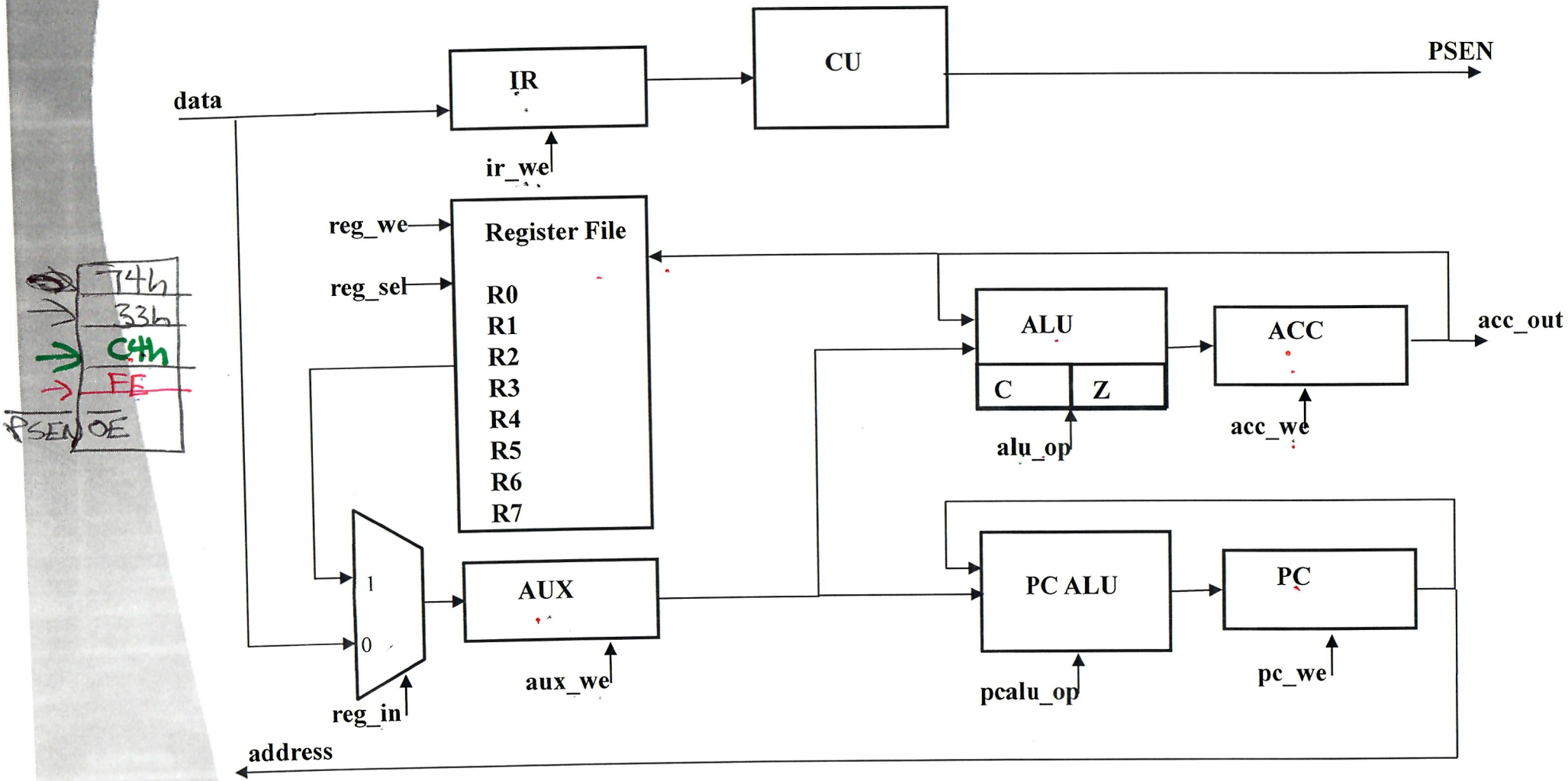
# WIMP51 microprocessor



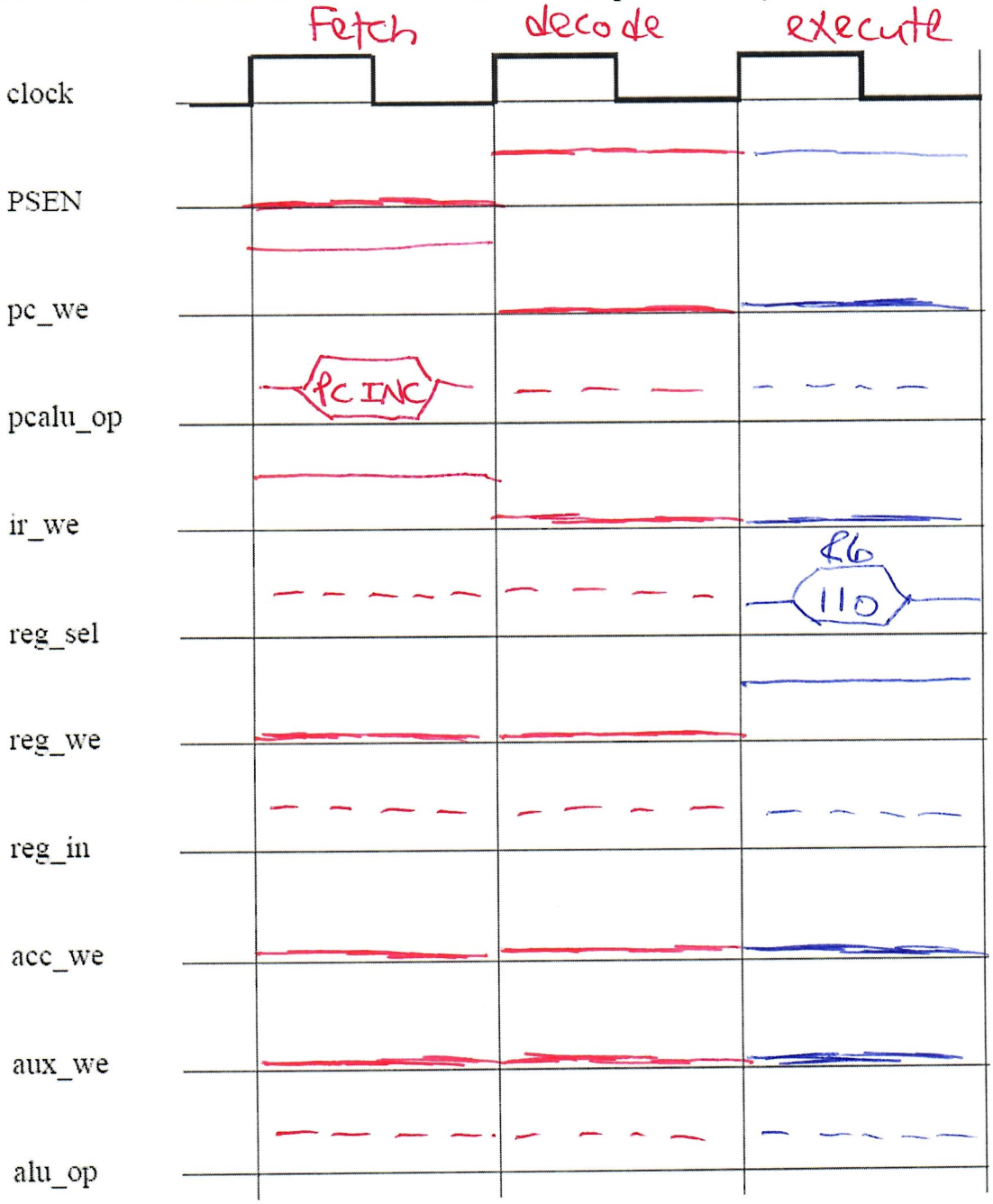
# Wimp51 Instructions

| | | | | |
|---|---|---|---|---|
| ✓ | MOV A, #D | 01110100 | dddddddd | A<=D |
| ✓ | ADDC A, #D | 00110100 | dddddddd | C,A<=A+D+C |
| | MOV Rn, A | 11111nnn | | Rn<=A |
| ✓ | MOV A, Rn | 11101nnn | | A<=Rn |
| ✓ | ADDC A, Rn | 00111nnn | | C,A<=A+Rn+C |
| ✓ | ORL A, Rn | 01001nnn | | A<=A or Rn |
| ✓ | ANL A, Rn | 01011nnn | | A<=A and Rn |
| ✓ | XRL A, Rn | 01101nnn | | A<=A xor Rn |
| | SWAP A | 11000100 | | $A<=A_{(3-0)}\&A_{(7-4}$ |
| | CLR C | 11000011 | | C<=0 |
| | SETB C | 11010011 | | C<=1 |
| ✓ | SJMP rel | 10000000 | aaaaaaaa | PC<=PC+rel+2 |
| ✓ | JZ rel | 01100000 | aaaaaaaa | PC<=PC+rel+2 if Z |

Draw the timing diagram for the following control signals on the diagram below when the WIMP51 is executing the instruction _____. Label the fetch, decode and execute parts of the instruction cycle on the clock portion of the signal below. MOV R6, A FEh

The dotted lines have been drawn as a graphical reference only and do not reflect the value of the corresponding signal. If you think a particular control signal is of no significance during a particular time period, indicate this opinion by using a **dashed** line to denote the signal as a "don't care" for that time period. Give "command" values for pcalu_op and alu_op; for example, indicate the value of "pcalu_op" as "PC_INC" or "PC_ADD" rather than representing this command by a numeric code. If you are not sure about a particular control signal, make an educated guess, indicate that the value you have marked is a guess, and explain why you made this particular guess. Remember, PSEN/ pulses low whenever a byte is being fetched from code memory.

**The WIMP51 architecture schematic is available in the notes provided to you for this exam.**



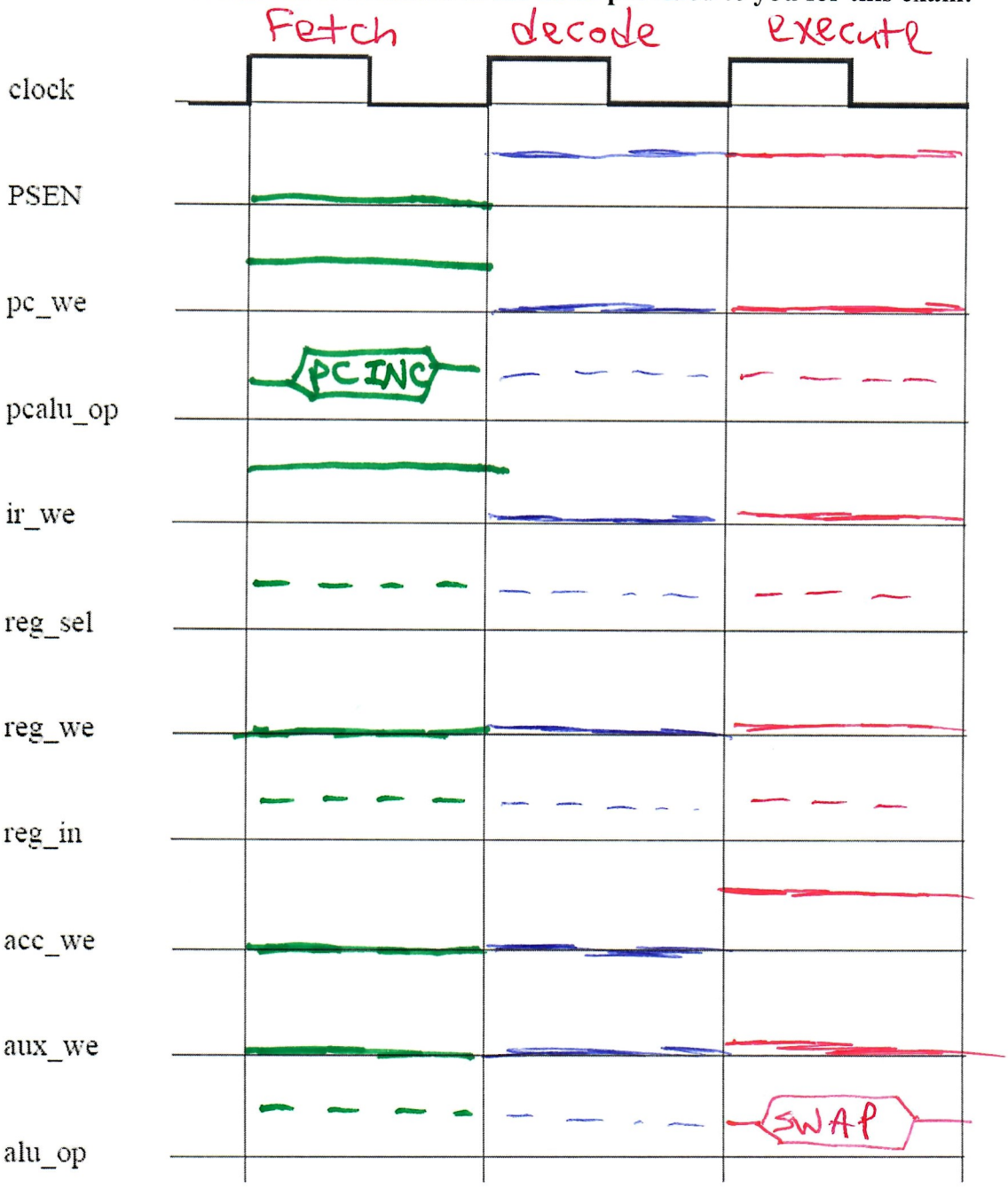

MOV R6, A

(read value)

Draw the timing diagram for the following control signals on the diagram below when the WIMP51 is executing the instruction _____. Label the fetch, decode and execute parts of the instruction cycle on the clock portion of the signal below. SWAP A

The dotted lines have been drawn as a graphical reference only and do not reflect the value of the corresponding signal. If you think a particular control signal is of no significance during a particular time period, indicate this opinion by using a **dashed** line to denote the signal as a "don't care" for that time period. Give "command" values for pcalu_op and alu_op; for example, indicate the value of "pcalu_op" as "PC_INC" or "PC_ADD" rather than representing this command by a numeric code. If you are not sure about a particular control signal, make an educated guess, indicate that the value you have marked is a guess, and explain why you made this particular guess. Remember, PSEN/ pulses low whenever a byte is being fetched from code memory.

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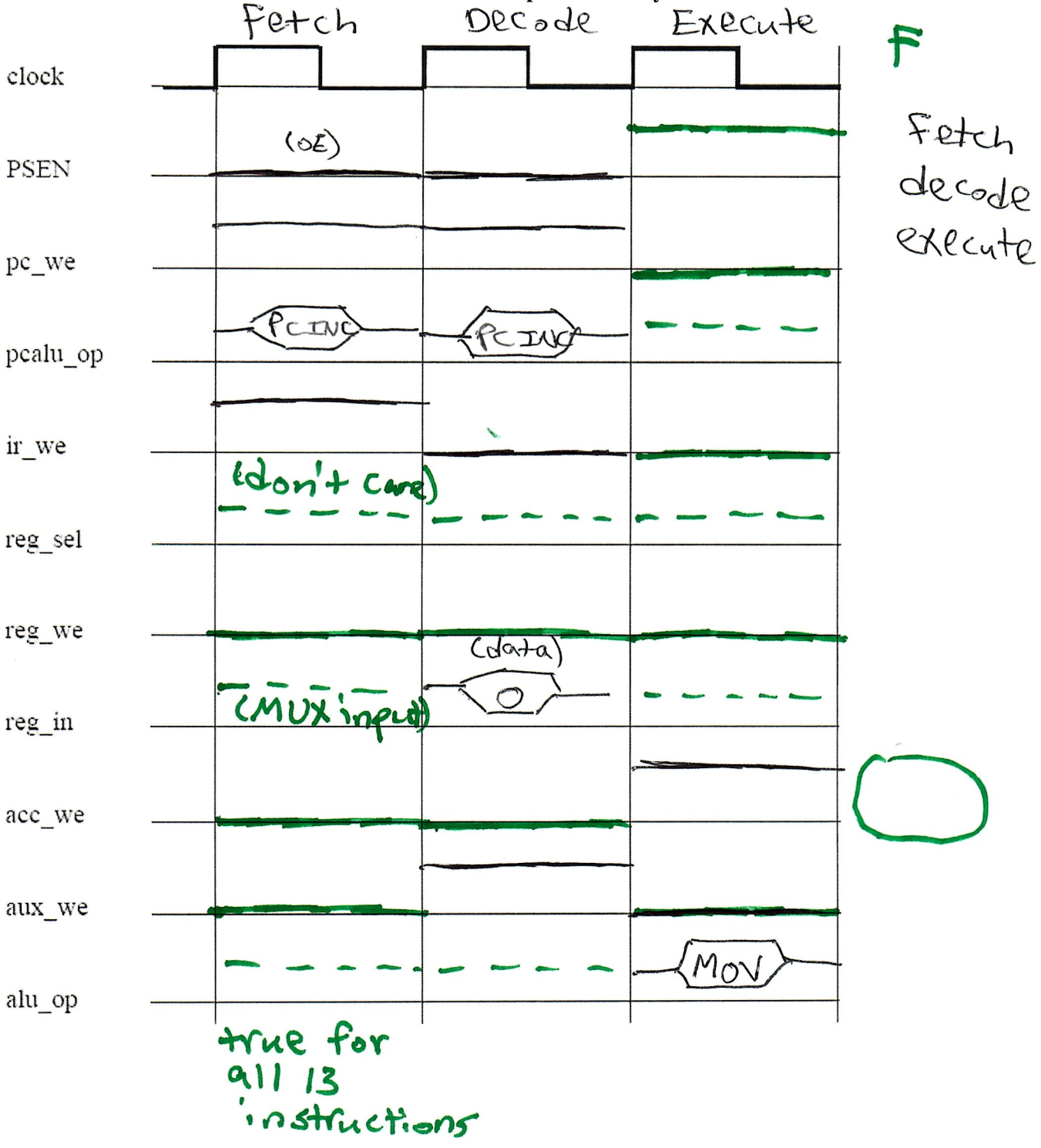


Draw the timing diagram for the following control signals on the diagram below when the WIMP51 is executing the instruction MOV A, #33h Label the fetch, decode and execute parts of the instruction cycle on the clock portion of the signal below.

MOV A, #33h 74h 33h

The dotted lines have been drawn as a graphical reference only and do not reflect the value of the corresponding signal. If you think a particular control signal is of no significance during a particular time period, indicate this opinion by using a **dashed** line to denote the signal as a "don't care" for that time period. Give "command" values for pcalu_op and alu_op; for example, indicate the value of "pcalu_op" as "PC_INC" or "PC_ADD" rather than representing this command by a numeric code. If you are not sure about a particular control signal, make an educated guess, indicate that the value you have marked is a guess, and explain why you made this particular guess. Remember, PSEN/ pulses low whenever a byte is being fetched from code memory.

**The WIMP51 architecture schematic is available in the notes provided to you for this exam.**



Draw the timing diagram for the following control signals on the diagram below when the WIMP51 is executing the instruction _____. Label the fetch, decode and execute parts of the instruction cycle on the clock portion of the signal below. ADDC A, #61h 34h 61h

The dotted lines have been drawn as a graphical reference only and do not reflect the value of the corresponding signal. If you think a particular control signal is of no significance during a particular time period, indicate this opinion by using a **dashed** line to denote the signal as a "don't care" for that time period. Give "command" values for pcalu_op and alu_op; for example, indicate the value of "pcalu_op" as "PC_INC" or "PC_ADD" rather than representing this command by a numeric code. If you are not sure about a particular control signal, make an educated guess, indicate that the value you have marked is a guess, and explain why you made this particular guess. Remember, PSEN/ pulses low whenever a byte is being fetched from code memory.

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